CSE331: Automata and Computability

Assignment 3

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Show the following language is not regular using pumping lemma

1. L = { $0^{2n}1^n$| n ≥ 0 }, where Σ = {0, 1}.

- Assuming : Suppose L is a regular language, then it must satisfy the pumping property.
- Choosing a String : $\omega = 0^{2s} 1^s$ $[2s + s \geq s]$
- Splitting : $\omega = x \cdot y \cdot z$ [where $|xy| \leq s$ and $|y| \geq 1$]

  $x = 0^p$; $y = 0^q$; $z = 0^{s-p-q} 0^s . 1^s$
- Checking if xy^iz belongs to L for all $i \geq 0$:

  $i = 0$: $xy^0z = xz = 0^p . 0^{s-p-q} . 0^s . 1^s = \cancel{0^p} . 0^s . \cancel{0^{-p}} . 0^{-q} . 0^s . 1^s = 0^{2s-q} . 1^s$

  Here, 0 occurs $2s-q$ times and 1 occurs s times. But 0 should occur 2n times (twice of 1s occurences). So, $0^{2s-q} . 1^s \notin L$, a contradiction. L is not regular.

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2. L = { w is not a palindrome }, where Σ = {0, 1}.

Sol^n:

If a language is not regular, its complement is also not regular. So we will first prove that $\overline{L}$ is not regular.

$$\overline{L} = \{ w \text{ is a palindrome} \}, \text{ where } \Sigma = \{0,1\}$$

- Suppose $\overline{L}$ is regular. Then it must satisfy the pumping property.
- $w = 0^s 1 0^s$ where $|w| \geq s$
  $2s+1 \geq s$
- $w = xyz$ where $|xy| \leq s$ and $|y| \geq 1$

$$\underset{0^p}{\underset{\downarrow}{x}}\ \underset{0^q}{\underset{\downarrow}{y}}\ \underset{0^{s-p-q}10^s}{\underset{\downarrow}{z}}$$

- Also xy^iz must belong to L for all $i \geq 0$.

$$\begin{aligned} i = 2,\ xyyz &= 0^p 0^q 0^q 0^{s-p-q} 1 0^s \\ &= \cancel{0^p}\cancel{0^q} 0^q 0^s \cancel{0^{-p}}\cancel{0^{-q}} 1 0^s \\ &= 0^q 0^s 1 0^s \\ &= 0^{q+s} 1 0^s \notin L \end{aligned}$$

- Contradiction! Hence L is not regular.

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3. L = { a^n where n is prime}, where Σ = {a}.

primes : { 2, 3, 5, 7, 11, ... }

L : { aa, aaa, aaaaa, ... }

- Assuming : L is a regular language and satisfies the pumping property.
- Choosing a string : $\omega = a^s$ $[s \geq s]$
- Splitting : $\omega = xyz$ [where $|xy| \leq s$ and $|y| \geq 1$]

  $x = a^p$, $y = a^q$, $z = a^{s-p-q}$
- Checking if $xy^iz \in L$ for all $i \geq 0$:

violates $|y| \geq 1$

$2 \times (0+1)$

$3 \times (0+1)$

$i=0$: $xy^0z = xz = a^p . a^{s-p-q} = a^{s-q}$?

$i=1$: $xy^1z = xyz = \cancel{a^p} . \cancel{a^q} . a^s . \cancel{a^{-p}} . \cancel{a^{-q}} = a^s$?

$i=s+1$: $xy^{(s+1)}z = xy^s . y^1 . z = \cancel{a^p} . a^{sq} . \cancel{a^q} . a^s . \cancel{a^{-p}} . \cancel{a^{-q}}$

$= a^{sq+s} = a^{s(q+1)} \notin L$

Here, the length of ω is $s \times (q+1)$. But a prime number is only divisible by 1 and itself. The length of ω is divisible by s and $(q+1)$, making it non-prime except for when $s =$ prime and $q = 0$, but q cannot be zero according to the condition in step 3, $|y| \geq 1$. Thus L is not regular.

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4. L = { $a^m b^n$ where $m \neq n$}, where Σ = {a, b}.

- Assuming : that L is regular and satisfies the pumping property
- Choosing ω : $\omega = a^s . b^{s+s!}$ $[s+s+s! \geq s]$
- Splitting : $\omega = xyz$ $[|xy| \leq s$ and $|y| \geq 1]$

  $x = a^p$, $y = a^q$, $z = a^{s-p-q} . b^{s+s!}$

- Checking if $xy^iz \in L$ for all $i \geq 0$:
  - $i=1$: $xyz = \cancel{a^p}.\cancel{a^q}.a^s.\cancel{a^{-p}}.\cancel{a^{-q}}.b^{s+s!} = a^s.b^{s+s!}$
  - $i=2$: $xyyz = \cancel{a^p}a^q\cancel{a^q}a^s\cancel{a^{-p}}\cancel{a^{-q}}b^{s+s!} = a^{s+q}.b^{s+s!}$

  - $i = \frac{s!}{q}+1$: $xy^{\left(\frac{s!+q}{q}\right)}z = a^p.(a^q)^{\left(\frac{s!+q}{q}\right)}.a^{s-p-q}.b^{s+s!}$

$$= a^p.a^{(s!+q)}.a^s.a^{-p}.a^{-q}.b^{s+s!}$$
$$= \cancel{a^p}.a^{s!}.\cancel{a^q}.a^s.\cancel{a^{-p}}.\cancel{a^{-q}}.b^{s+s!}$$
$$= a^{s+s!}.b^{s+s!} \notin L$$

Contradiction, when $i = \frac{s!}{q}+1$, ω has equal number of a_s and b_s.
L is not a regular language.

Context free grammar

1. L = { w ∈ {0,1}* : w = 0^n1^n, where n is odd }

$S \rightarrow 0P1$
$P \rightarrow 0S1 \mid \epsilon$

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2. L = { w ∈ {0,1}* : the length of w is divisible by 3 }

$S \rightarrow PSPP \mid \epsilon$
$P \rightarrow 0 \mid 1$

3. L = { w ∈ {0,1}* : $w_1\#w_2$, the number of 0's in w_1 is equal to number of 1's in w_2}

$S \rightarrow 1S \mid S0 \mid 0S1 \mid \#$

4. L = { w ∈ {0,1,2}* : w = $0^i2^j1^k$, where i = k, i,k ≥ 1 and j ≥ 2 }

$$w = 0^i\, 2^j\, 1^k \qquad i,k \geq 1,\quad j \geq 2$$

$$= \underbrace{0^k\, 2^j\, 1^k}$$

$$S \rightarrow 0P1$$

$$P \rightarrow 0P1 \mid y$$

$$y \rightarrow 22x$$

$$x \rightarrow 2x \mid \epsilon$$

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5. L = { w ∈ {0,1,2}* : w = $0^i 2^j 1^k$, where i is multiple of two, k is two more than multiple of 3, j = k+i and i,j,k ≥ 0 }

$$w = 0^i\, 2^j\, 1^k \qquad j = k+i$$

$$= 0^{2i}\, 2^{3k+2i+2}\, 1^{3k+2} \qquad = 3k+2+2i$$

$$= 2i+3k+2$$

$$\cancel{= \underbrace{0^{2i}\, 2^i}\, \underbrace{2^k\, 1^{3k}}\, 1^2} \qquad 0^{2i}\, 2^{2i}\, 2^2\, 2^{3k}\, 1^{3k}\, 1^2$$

$$S \rightarrow xy$$

$$x \rightarrow 00x22 \mid 22$$

$$y \rightarrow 222y111 \mid 11$$

6. L = { w ∈ {0,1}* : the number of 0s and 1s are different in w }

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$L_7 = \{ w \in \{0,1\}^* : \text{The number of 0's and 1's different in } w \}$

First Solve the CFG for equal 0's and 1's

$S \rightarrow 0S1 \mid 1S0 \mid \underline{SS} \mid \varepsilon$

$\rightarrow$ For handling 0110

A B

$\boxed{0110}\boxed{110010}$

Both blocks Contain equal amount of 0's and 1's. To make it unequal I can insert 0's before block A, in middle of A and B and after block B.

$T \rightarrow S0S$

$S \rightarrow 0S1 \mid 1S0 \mid SS \mid 0S \mid \varepsilon$

There Could also be more 1's than 0's.

$S \rightarrow A \mid B$

$A \rightarrow X0X$

$X \rightarrow 0X1 \mid 1X0 \mid XX \mid 0X \mid \varepsilon$

$B \rightarrow Y1Y$

$Y \rightarrow 0Y1 \mid 1Y0 \mid YY \mid 1Y \mid \varepsilon$

Regular expression to CFG

(a*b* + (ac + b*c)b)*

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$$(\underbrace{\overbrace{a^*}^{P}\overbrace{b}^{Q}}_{X} + \underbrace{(\overbrace{\overbrace{ac}^{E} + \overbrace{\underbrace{b^*}_{G}\underbrace{c}_{H}}^{F}}^{C})\overbrace{b}^{D}}_{Y})^*$$

$S \rightarrow RS \mid \varepsilon$

$R \rightarrow X \mid Y$

$X \rightarrow PQ$

$P \rightarrow aP \mid \varepsilon$

$Q \rightarrow b$

$Y \rightarrow CD$

$D \rightarrow b$

$C \rightarrow E \mid F$

$E \rightarrow ac$

$F \rightarrow GH$

$H \rightarrow c$

$G \rightarrow bG \mid \varepsilon$

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Derivation, Parse tree and Ambiguity

Let $\Sigma = \{0, 1\}$. Consider the following grammar over Σ.

$$S \to 0S0 \mid 1S1 \mid A$$
$$A \to 00A \mid 01A \mid 10A \mid 11A \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string 01101010. (3 points)

(b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the given string in (a). (4 points)

(d) How many four-letter strings will have exactly one parse tree in the given grammar? (1 point)

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<table>
<tr>
<td></td>
<td>S
⇒A
⇒01A
⇒0110A
⇒011010A
⇒01101010A
⇒01101010</td>
</tr>
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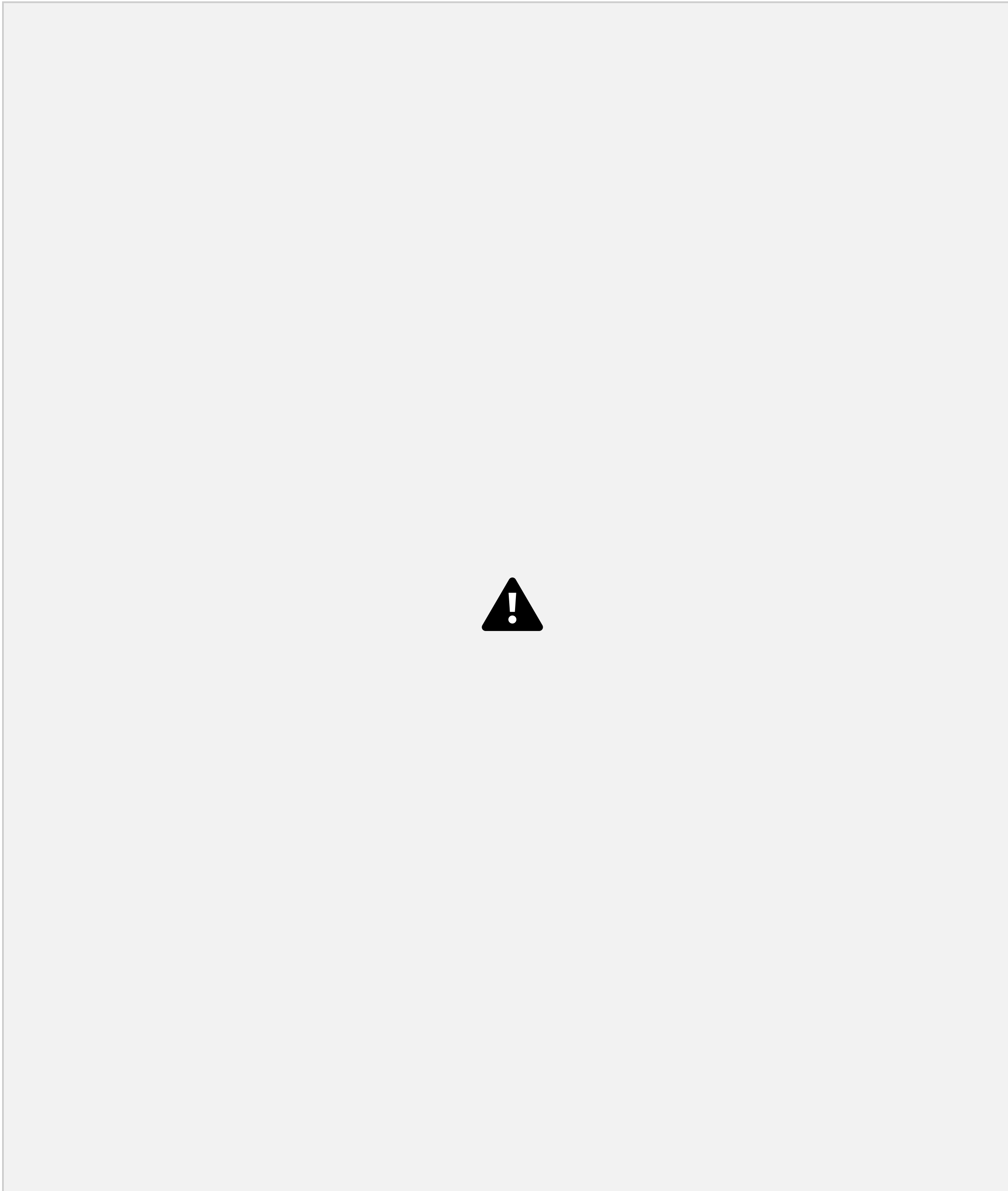
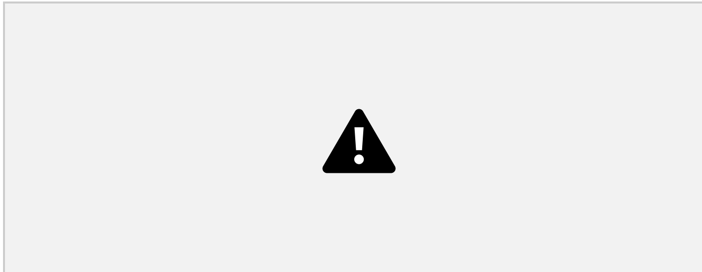
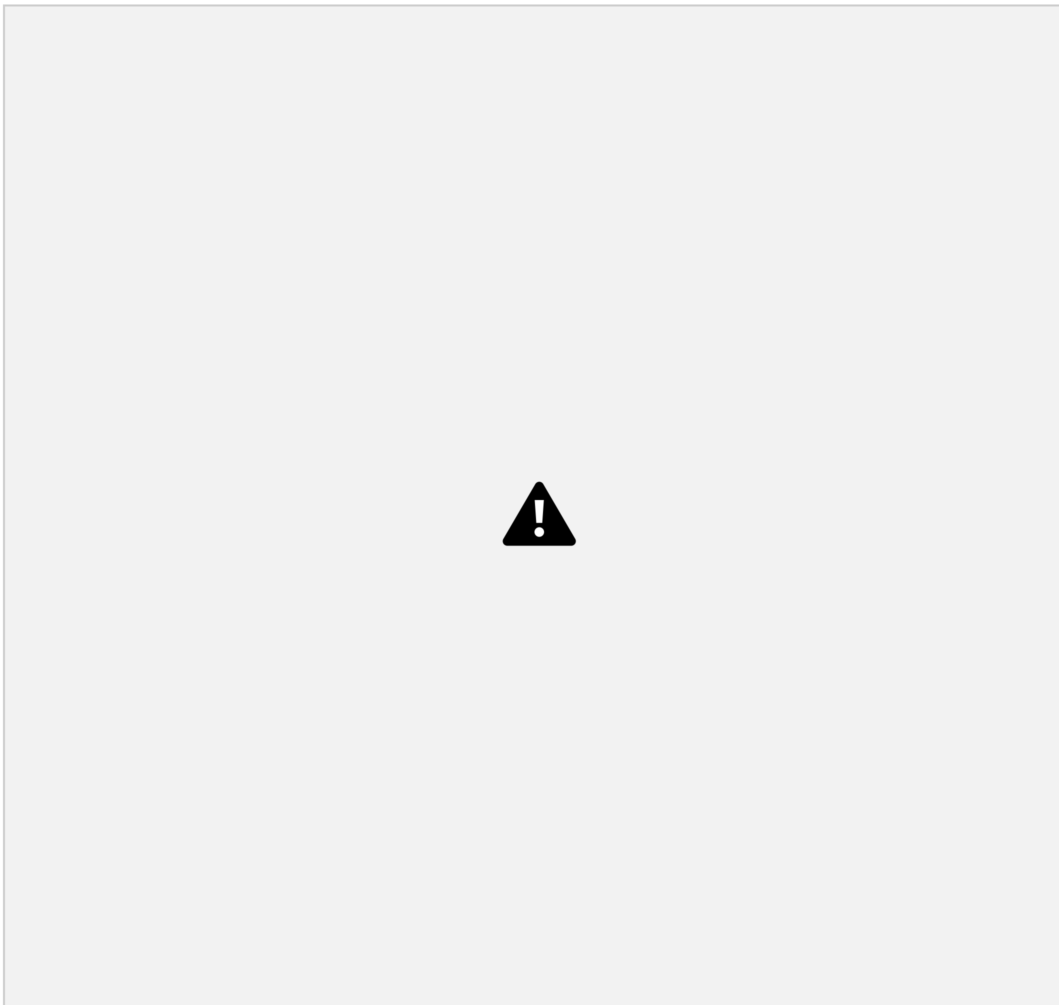


Push down automata

1. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k$, where $i = k$, $i,k \geq 1$ and $j \geq 2 \}$

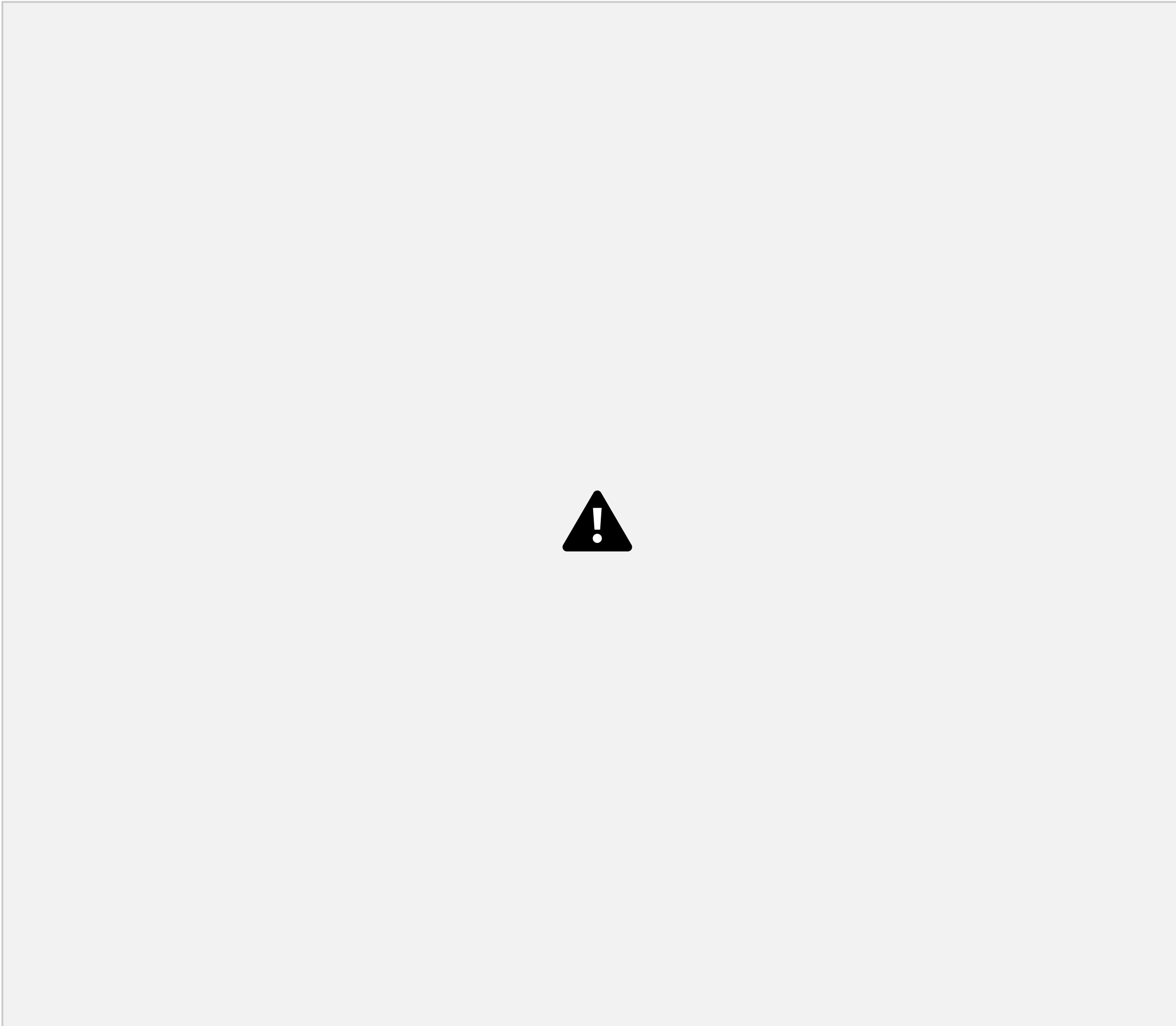


2. L = { w ∈ {0,1,2}* : w = $0^i 2^j 1^k$, where i is multiple of two, k is two more than multiple of 3, j = k+i and i,j,k ≥ 0 }

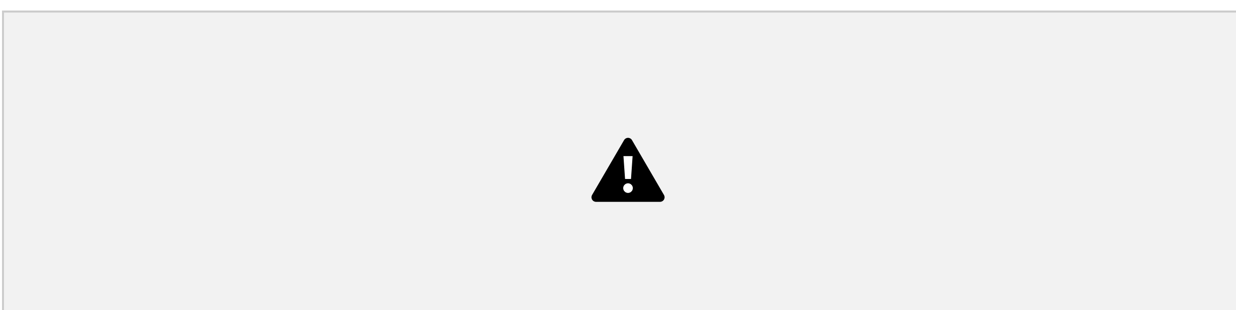


3. L = { w ∈ {0,1}* : the number of 0s and 1s are different in w }

First do the PDA for equal amount of 0 and 1 to get an idea about the language given



4. L = {w#x: w,x ∈ {0,1}* and w^R is a substring of x}



5. Recall that for a string w, |w| denotes the length of w. Σ = {0,1}

L1 = {w ∈ Σ* : w contains exactly two 1s}

L2 = {x#y : x ∈ Σ*, y ∈ L1, |x| = |y|}

Construct a PDA for L2.

Turing Machine

Problem 1

Consider the following Turing Machine M with input alphabet $\Sigma = \{a, b\}$. The reject state $q_{rej}$is not shown, and if from a state there is no transition on some symbol then

1. Give the formal definition of M as a tuple.
2. Describe each step of the computation of M on the input baabab as a sequence of instantaneous descriptions.
3. Describe the language recognized by M. Give an informal argument that outlines the intuition behind the algorithm used by M justifies your answer.

You can show one transition function from the many we have in the diagram.

Problem 2

For $\Sigma = \{\#, a, b\}$, design a Turing machine to recognize the language $L = \{a^{2n}\#b^n \mid n \geq 0\}$

Problem 3

For $\Sigma = \{a, b\}$, design a Turing machine to recognize the language $L = \{0^n\#1^n\#2^m \mid n, m \geq 0$ and $n < m\}$

Problem 4

Prove that the following languages are decidable.
$A_{DFA}\{\langle D, w\rangle \mid D$ is a DFA that accepts input string $w\}$
Check the notes and quiz solution, try to give a high level description of how your machine will work.